

Question 1:

(12 marks)

- (a) Express 0.04678 in scientific notation correct to 3 significant figures. [2]
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- (b) Rationalise the denominator of $\frac{3+\sqrt{2}}{2-\sqrt{2}}$. [3]

Simplify your answer as far as possible.

- (c) Solve for θ if $2\cos\theta=1$ and $0^\circ \leq \theta \leq 360^\circ$. [3]

- (d) Solve for k if $3(k-1)-2(k-2)>1$. [2]

- (e) Express the recurring decimal $0.\dot{2}\dot{2}$ as a fraction in its lowest terms. [2]

Question 2: *Start a new sheet of paper*

(12 marks)

- (a) Find the derivatives of each of the following with respect to x :

(i) $x^3 - 5x^2 + 7x + e^3$

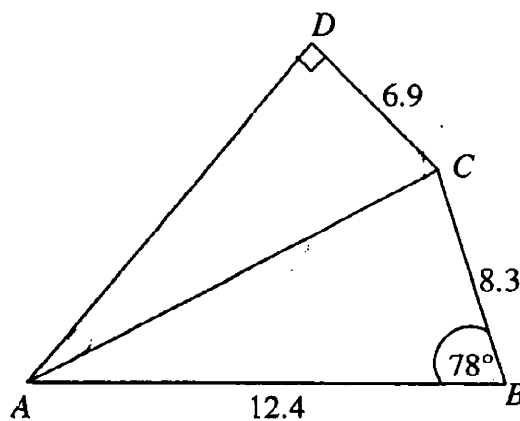
(ii) $\ln(x^2 + 5)$

(iii) $\tan(3x)$

(iv) $\frac{\sin x}{x^3}$

[6]

- (b) The quadrilateral $ABCD$ is shown in the diagram – *all measurements are in centimetres*.



(Diagram not to scale)

- (i) Find, using the Cosine Rule, or otherwise, the length of AC , correct to one decimal place.
- (ii) Find the size of $\angle CAD$, giving your answer to the nearest degree.
- (iii) Calculate, correct to one decimal place the area of quadrilateral $ABCD$.

[6]

Question 3: *Start a new sheet of paper*

(12 marks)

A circle C , with centre $(-4, 2)$ is tangent to the straight line l with the equation $3x + 4y - 16 = 0$. The perpendicular distance from the centre of this circle to the straight line l is equal to the radius of the circle.

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- (a) Find the x and y intercepts, A and B , respectively of the line l . [2]
- (b) Show that the radius of the circle, C , is 4 units. [3]
- (c) Write down the equation of the circle, C , in the form
- $$(x - h)^2 + (y - k)^2 = r^2 \quad [1]$$
- (d) On a suitable number plane, sketch the circle, C and the straight line, l . [2]
- (e) Verify that the coordinates of the point of intersection, E , of the circle, C , and the line, l is $(-1.6, 5.2)$. [1]
- (f) Find the equation of the line, k , that passes through $(4, -2)$, and is also perpendicular to the line l . [2]
- (g) Verify that the circle, C , intersects the y axis at the point D , with coordinates $(0, 2)$. [1]

Question 4: *Start a new sheet of paper* (12 marks)

- (a) (i) Find the number of terms there are in the arithmetic series

$$258 + 251 + 244 + \dots - 288 - 295 ?$$

- (ii) Calculate the sum of the arithmetic series in (a)(i) above. [4]

- (b) (i) Find the discriminant of the quadratic equation

$$x^2 - (2 - k)x + 1 = 0.$$

- (ii) Hence, or otherwise, determine the values of k for which
the graph of

$$y = x^2 - (2 - k)x + 1$$

cuts the x – axis in two distinct points. [4]

- (c) A parabola's equation is given by $y = x^2 - 2x - 3$.

- (i) Find the focal length.
(ii) Determine the coordinates of the vertex V .
(iii) Find the coordinates of the focus S . [4]

Question 5: *Start a new sheet of paper*

(12 marks)

- (a) The common ratio r of a geometric progression satisfies the quadratic equation

$$2r^2 - 3r - 2 = 0.$$

- (i) Solve for r

If the sum to infinity of the same progression is 6,

- (ii) Explain why, in this case, r can only take on one value.
Hence, state the common ratio r .

- (iii) Show that the first term a of this progression, is 9. [5]

- (b) A population of eels in a certain river has become too large to be supported by its environment and it decreases from time $t = 0$ to a stable level.

The size of the population in thousands of eels is given by $g(t)$, where t is measured in weeks.

The rate of change of the population is given by $g'(t) = -5e^{-2t}$ for $t \geq 0$. It is known that when $t = 0$, $g = 10$.

- (i) Find, by integrating $g'(t) = -5e^{-2t}$, the size of the population of eels at any time t .
- (ii) Sketch the graph of $y = g(t)$ on a number plane.
- (iii) State the long-term stability level of the eel population in the river. [7]

Question 6: *Start a new sheet of paper*

(12 marks)

(a) Consider the function $f(x) = x^3 - x^2 - x + 1$.

-
- (i) By using the *grouping in pairs method for factorising*, or *otherwise*, factorise fully the expression $x^3 - x^2 - x + 1$.
- (ii) Evaluate $f(1)$ **and** $f(-1)$.
- (iii) Hence, or otherwise, state the x intercepts of $f(x)$.
- (iv) Determine the coordinates **and** the nature of the stationary points on $y = f(x)$.
- (v) Find the coordinates of the point(s) of inflexion of $y = f(x)$.
- (vi) Hence, sketch the graph of $y = f(x)$ on a number plane, showing all essential features. [8]

(b) Forty five percent of a population are of blood group O , 40% are of blood group A and the remainder are neither group O nor group A .

Three people are chosen at random from the population.

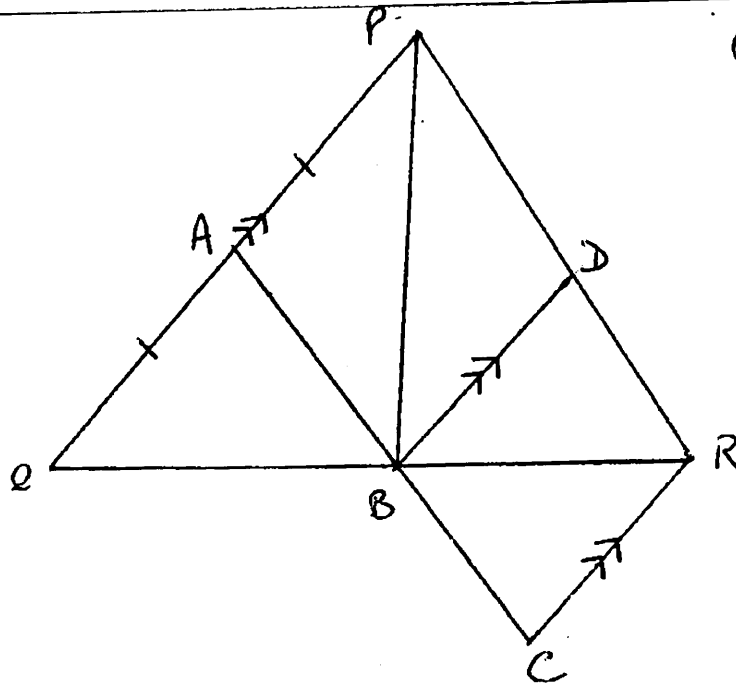
By drawing a tree diagram, or otherwise, find the probability that

- (i) all three people are of blood group A ;
- (ii) two of the people are of blood group A and the other is group O ;
- (iii) there is one person each of group O , group A and neither group O nor group A . [4]

Question 8: *Start a new sheet of paper*

(12 marks)

In triangle PQR , B is a point on QR such that PB is the angle bisector of angle RPQ . A is the midpoint of QP . The straight line joining AB is produced to a point C such that $CR \parallel QP$.



(Diagram is not to scale)

Copy this diagram onto your answer sheet.

- (a) Prove, with reasons, that $\triangle BAQ$ is similar to $\triangle BCR$. [3]

- (b) Hence, state why $\frac{RC}{QA} = \frac{RB}{QB}$. [1]

- (c) BD is a line drawn parallel to QP meeting PR at D .

Show that $\frac{RD}{PD} = \frac{RB}{QB}$. [2]

- (d) By showing $\triangle BRD$ is similar to $\triangle QRP$, or otherwise, show that $\frac{RD}{PD} = \frac{PR}{PQ}$. [3]

- (e) Hence, or otherwise, show that $RC = \frac{1}{2}PR$. [3]

Question 7: *Start a new sheet of paper*

(12 marks)

(a) (i) On a suitable number plane sketch the graph of $y = |x - 1|$.

(ii) Solve the inequality $|x - 1| < 1$.

(iii) Hence, or otherwise, evaluate $\int_0^2 |x - 1| dx$. [4]

(b) (i) Find $\int \frac{1}{2x - 5} dx$.

(ii) Find $\int_0^{\frac{\pi}{3}} \cos(6x) dx$, leaving your answer in simplified form. [4]

(c) Let A be the area enclosed by the curve $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, the x axis and the lines $x = 4$ and $x = 9$.

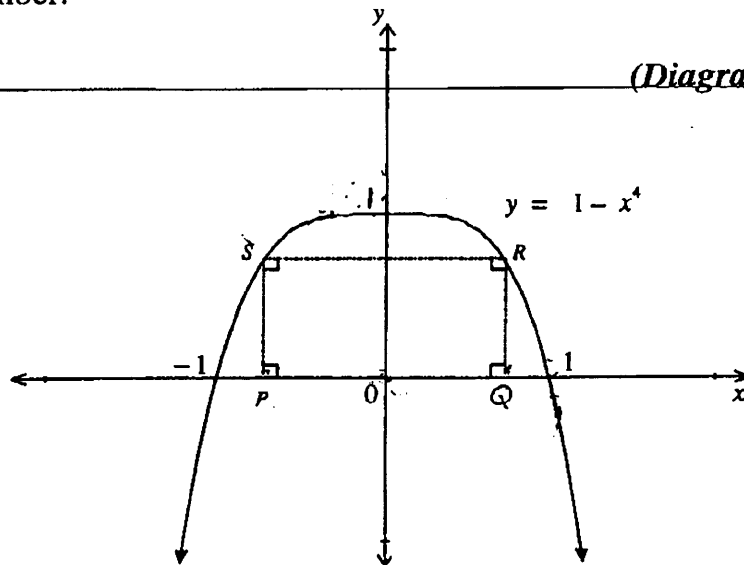
Find the volume of the solid formed by rotating this area through 360° about the x axis.

Leave your answer in simplified form and in terms of π . [4]

Question 9: *Start a new sheet of paper*

(12 marks)

The diagram shows part of the graph of the function $y = 1 - x^4$, where x is any real number.



- (a) Calculate I , the area enclosed between the graph of $y = 1 - x^4$ and the x axis. [3]

- (b) The rectangle $PQRS$ has coordinates

$$P(-u, 0), Q(u, 0), R(u, y(u)), \text{ and } S(-u, y(-u))$$

where $0 < u < 1$.

Let the area of the rectangle $PQRS$ be denoted by $A(u)$.

- (i) State why $A(u) = 2u(1 - u^4)$ square units.
- (ii) Verify that $A(0.5) = \frac{15}{16}$.
- (iii) Let u_1 be the value of u for which $A(u)$ is a maximum. Find the value of u_1 correct to five significant figures.
- (iv) Calculate the maximum value of $A(u)$, giving your answer correct to 5 significant figures.
- (v) Show that if $A(u) = uI$, then $u = u_1$. [9]

Question 10: *Start a new sheet of paper*

(12 marks)

Dusty Lifesaver is retiring from the work force. Dusty has a lump sum (from a superannuation fund) of $\$P$ to invest. Dusty is advised that it will take $\$m$ to cover living expenses in any given year after retirement with the remainder invested at $r\%$ per annum and compounded annually. The living expense of $\$m$ is taken out of the remainder at the beginning of each year.

Let $R = 1 + \frac{r}{100}$ and assuming $P > R(P - m)$ is satisfied.

(a) Show that

(i) after 1 year Dusty has $\$R(P - m)$ remaining.

(ii) after 2 years Dusty has $\$PR^2 - mR(1 + R)$ remaining.

(iii) after 3 years Dusty has $\$PR^3 - mR(1 + R + R^2)$ remaining. [3]

(b) Show that $1 + R + R^2 = \frac{R^3 - 1}{R - 1}$. [1]

(c) By first observing the pattern established in (a), show that after n years the remaining superannuation in dollars is given by

$$PR^n - \frac{mR(R^n - 1)}{R - 1}. \quad [2]$$

(d) By first equating the expression in (c) to zero, show that Dusty's superannuation is *exhausted* when

$$R^n = \frac{mR}{P - R(P - m)}. \quad [3]$$

(a) Hence, or otherwise, find the number of years Dusty can survive if $P = \$400\,000$, $m = \$40\,000$ and $r = 7\%$ per annum. [3]

END OF EXAMINATION

Question 1

a) 4.68×10^{-2}

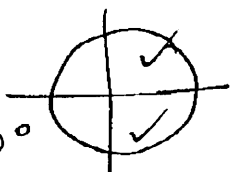
(2)

b) $\frac{3+\sqrt{2}}{2-\sqrt{2}} = \frac{(3+\sqrt{2})(2+\sqrt{2})}{(2-\sqrt{2})(2+\sqrt{2})}$
 $= \frac{6+5\sqrt{2}+2}{4-2}$
 $= \frac{8+5\sqrt{2}}{2}$

(3)

c) $2\cos\theta = 1 \quad 0^\circ \leq \theta \leq 360^\circ$
 $\Rightarrow \cos\theta = \frac{1}{2}$

$\Rightarrow \theta = 60^\circ \text{ or } 300^\circ$



(3)

d) $\Rightarrow 3(k-1) - 2(k-2) > 1$

$\Rightarrow 3k-3-2k+4 > 1$

$\Rightarrow k+1 > 1$

$\Rightarrow k > 0$

(2)

e) $\frac{0.\dot{2}\dot{2}}{99} = \frac{22}{99}$
 $= \frac{2}{9}$

(2)

Question 2

a) i) $3x^2 - 10x + 7$

ii) $\frac{2x}{x^2+5}$

iii) $3\sec^2 3x$

iv) $\frac{u'v - uv'}{v^2} = \frac{\cos x \cdot x^3 - \sin x \cdot 3x^2}{x^6}$

$= \frac{x \cos x - 3 \sin x}{x^4}$

$x \neq 0$

(6)

b) i) $AC^2 = 12.4^2 + 8.3^2 - 2 \times 12.4 \times 8.3 \times \cos 78^\circ$
 $= 179.853 \quad (3 \text{ dp}) \quad \text{COS RULE}$

$\Rightarrow AC = \underline{13.4 \text{ correct to 1 dp}}$

ii) $\angle CAD = \sin^{-1} \frac{6.9}{AC}$

$= \underline{31^\circ} \text{ to nearest degree}$

iii) $\text{Area} = \text{Area } \triangle ADC + \triangle ACB$

$= \frac{1}{2} AC \times CD \sin \angle DCA$
 $+ \frac{1}{2} AB \times BC \times \sin 78^\circ$

$= \frac{1}{2} \times 13.41 \times 6.9 \times \sin 59^\circ$

$+ \frac{1}{2} \times 12.4 \times 8.3 \times \sin 78^\circ$

$= 36.66 + 50.34$

$= \underline{90.0 \text{ cm}^2} \quad (1 \text{ dp})$

(6)

Question 3

a) $l: 3x + 4y - 16 = 0$

x intercept $(\frac{16}{3}, 0)$

y intercept $(0, 4)$

②

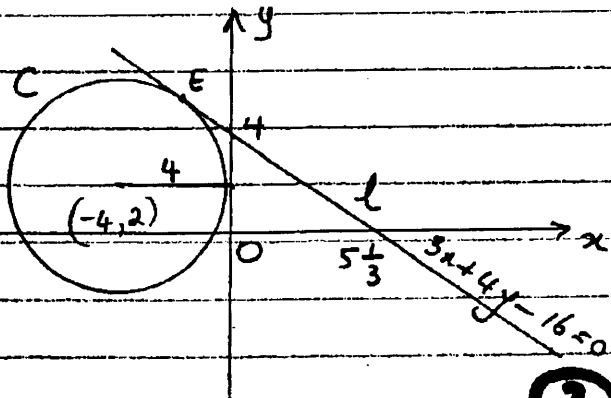
b) $r = d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|3(-4) + 4(2) - 16|}{\sqrt{3^2 + 4^2}}$
 $= \frac{20}{5} = 4$

③

c) $(x+4)^2 + (y-2)^2 = 4^2$

①

d)



②

e) $E(-1.6, 5.2)$ sub in

$l: 3(-1.6) + 4(5.2) - 16$

$= -4.8 + 20.8 - 16$

$= 0 \checkmark$ lies on line l .

$C: (-1.6+4)^2 + (5.2-2)^2$

$= 2.4^2 + 3.2^2$

$= 16 = r^2 \checkmark$ lies on circle

$\therefore E$ is on both $\therefore E$ is the intersection.

①

f) $m_l = -\frac{3}{4}$

$M_k = \frac{4}{3} \quad (4, -2)$

$y + 2 = \frac{4}{3}(x - 4)$

$3y + 6 = 4x - 16$

$4x - 3y - 22 = 0$

②

g) $(0, 2)$ lies on the x-axis.

sub $(0, 2)$ in C :

$(0+4)^2 + (2-2)^2 = 16 + 0$

$= 16 \checkmark$ lies on C

$\therefore (0, 2)$ is intersection D

①

Question 4

a) i) $a = 258, d = -7$

$T_n = -295 = a + (n-1)d$

$= 258 + (n-1)(-7)$

$= 258 - 7n + 7$

$\Rightarrow 7n = 560$

$\Rightarrow n = 80$

ii) $S_n = \frac{n}{2}(a + l)$

$= \frac{80}{2}(258 - 295)$

$= -1480$

④

b) i) $\Delta = b^2 - 4ac$

$= (-2-k)^2 - 4(1)(1)$

$= 4 - 4k + k^2 - 4$

$= k(-4 + k)$

ii) $\Delta > 0$ for 2 distinct pts

i.e. $k(k-4) > 0$

i.e. $k < 0$ and $k > 4$

④

c) i) $y = x^2 - 2x - 3$
 $= (x-1)^2 - 1 - 3$
 $\Rightarrow y+4 = (x-1)^2$
 $|a| = \frac{1}{4}$

ii) $V = (1, -4)$

iii) $S = (1, -3\frac{3}{4})$

(4)

Question 5

i) $2r^2 - 3r - 2 = 0$

$\Rightarrow (2r+1)(r-2) = 0$

$r = -\frac{1}{2}$ or $r = 2$

If the sequence has a sum to infinity
 $|r| < 1$ i.e. $r = -\frac{1}{2}$

i) $S_{\infty} = \frac{a}{1-r} = 6$

$a = 6 \times (1 - -\frac{1}{2})$

$a = 9$

(5)

b) i) $g'(t) = -5e^{-2t}$
 $\Rightarrow g(t) = \frac{5}{2}e^{-2t} + C$

$g(0) = 10$

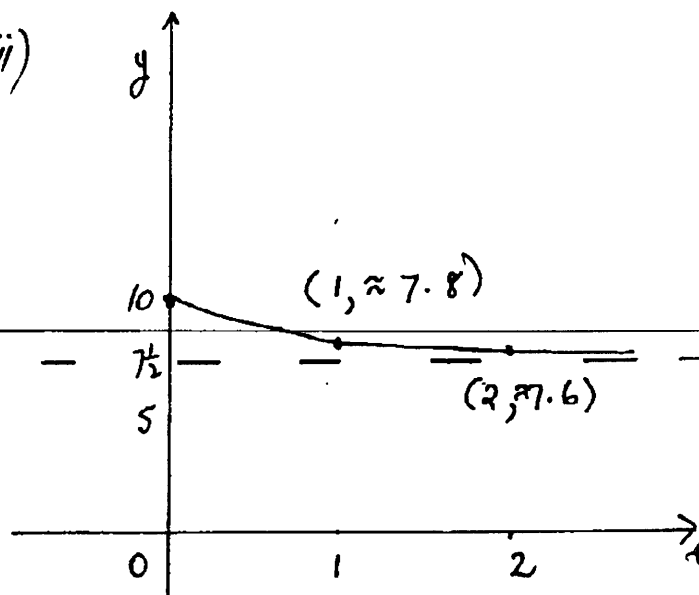
$\Rightarrow 10 = \frac{5}{2}e^{-2 \cdot 0} + C$

$\Rightarrow 10 = 2\frac{1}{2} + C$

$\Rightarrow C = 7\frac{1}{2}$

$\Rightarrow g = \frac{5}{2}e^{-2t} + 7\frac{1}{2}$

ii)



iii) 7500 eels

(7)

Question 6

a) i) $x^2(x-1) - 1(x-1)$
 $= (x^2-1)(x-1)$

$= (x-1)(x+1)(x-1)$

ii) $f(1) = 0$, $f(-1) = 0$

iii) $(1, 0)$, $(-1, 0)$

iv) For stationary pt
 $f'(x) = 0$

1. $3x^2 - 2x - 1 = 0$

2. $(3x+1)(x-1) = 0$

$\Rightarrow x = -\frac{1}{3}$ or $x = 1$

$f''(x) = 6x - 2$

$x = -\frac{1}{3}$ $f''(x) < 0$ max

$x = 1$ $f''(x) > 0$ min.

ctd

MAX at $(-\frac{1}{3}, \frac{32}{27})$

MIN at $(1, 0)$

v) Now stationary inflection of

$$f''(x) = 0 \text{ (at } f'(x) \neq 0)$$

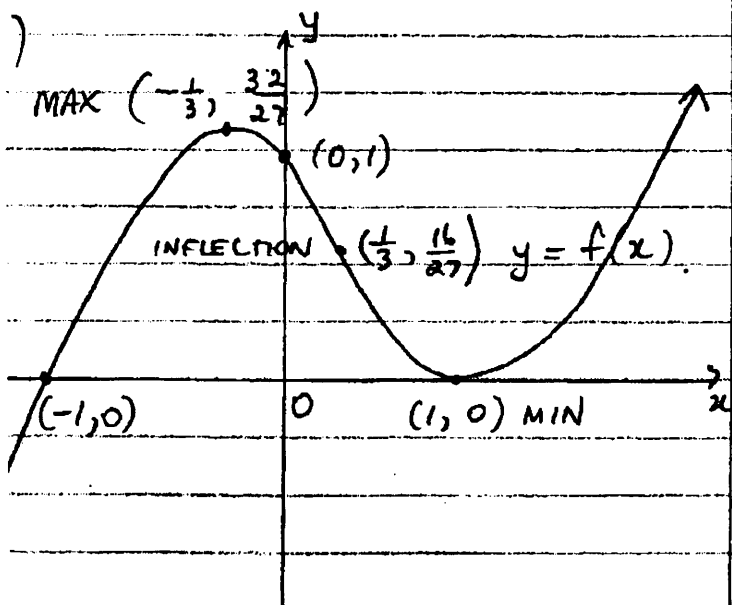
$$\text{i.e. } 6x - 2 = 0$$

$$\text{i.e. } x = \frac{1}{3}$$

$$\Rightarrow y = \frac{16}{27}$$

Coords $(\frac{1}{3}, \frac{16}{27})$

8



$$i) P(0) = 45\% \quad P(A) = 40\%$$

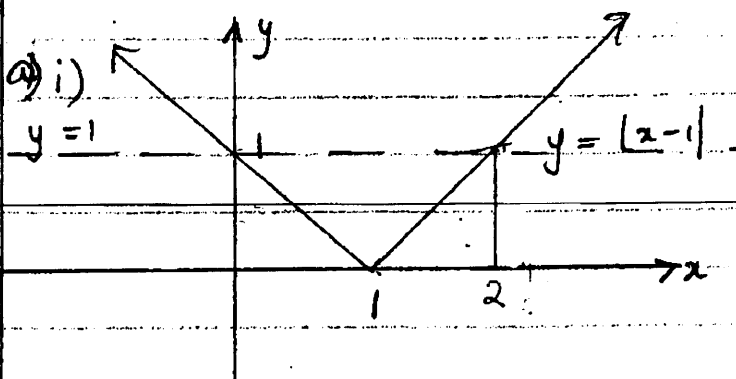
$$i) P = (0.4)^3 = 0.064$$

$$ii) P = 3 \times (0.4)^2 \times 0.45 = 0.216$$

$$iii) P = 6 \times 0.4 \times 0.45 \times 0.15 = 0.162$$

4

Question 7



$$ii) 0 < x < 2$$

$$iii) \int_0^2 |x-1| dx = \text{area under } y = |x-1| \text{ for } 0 < x < 2$$

$$= 2 \times \frac{1}{2} \times 1 \times 1$$

$$= 1$$

4

b)

$$i) \int \frac{1}{2x-5} dx = \frac{1}{2} \ln|2x-5| + C$$

$$ii) \int_0^{\pi/3} \cos 6x dx = \left[\frac{\sin 6x}{6} \right]_0^{\pi/3}$$

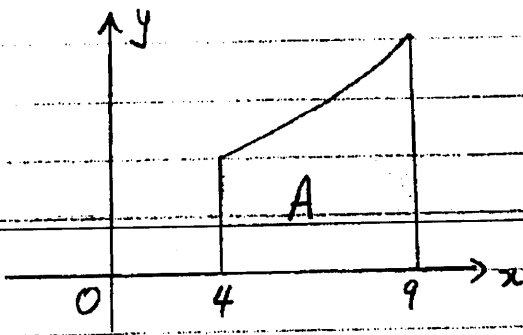
$$= \frac{\sin 2\pi}{6} - \frac{\sin 0}{6}$$

$$= 0$$

4

7 ctd

c)



$$\begin{aligned}
 V &= \pi \int_a^b y^2 dx \\
 &= \pi \int_4^9 \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx \\
 &= \pi \int_4^9 \left(x + 2 + \frac{1}{x} \right) dx \\
 &= \pi \left(\frac{x^2}{2} + 2x + \ln x \right) \Big|_4^9
 \end{aligned}$$

$$\begin{aligned}
 &= \pi \left(\frac{81}{2} + 18 + \ln 9 \right. \\
 &\quad \left. - \frac{4^2}{2} - 8 - \ln 4 \right)
 \end{aligned}$$

$$= \pi \left(\frac{85}{2} + \ln \frac{9}{4} \right)$$

$$\text{OR } \pi \left(\frac{85}{2} + 2 \ln \frac{3}{2} \right)$$

Question 8

a) In $\triangle BAP$, $\triangle BCR$

$\angle PAB = \angle RCB$ (alt $\angle$'s $PA \parallel CR$)

③ $\angle PBA = \angle RBC$ (vert opp $\angle$'s)

$\Rightarrow \triangle BAP \parallel \triangle BCR$ (equiangular)

b) corresponding sides of

① similar $\triangle$'s are in proportion

c) $BD \parallel QP \Rightarrow$

② $\frac{RD}{PD} = \frac{RB}{QB}$ (intercept theorem)

d) In $\triangle BCR$, $\triangle QRP$

$\angle R$ is common

$\angle DBR = \angle PQR$ (corr $\angle$'s $QP \parallel BD$)

$\Rightarrow \triangle BCR \parallel \triangle QRP$

$\Rightarrow \frac{RD}{BD} = \frac{RP}{QP}$ (corresponding sides)

But $\angle DBP = \angle APB$ (alt $\angle$'s $AP \parallel B$)
and $\angle DPB = \angle APB$ (given)

$\Rightarrow \angle DBP = \angle DPB$

$\Rightarrow \triangle PBD$ is isosceles

$\Rightarrow BD = PD$

$\therefore \frac{RD}{PD} = \frac{RP}{QP}$ from * ③

e) Proven $\frac{RC}{QA} = \frac{RB}{QB}$, $\frac{RD}{PD} = \frac{RB}{QB}$

$\Rightarrow \frac{RC}{QA} = \frac{RD}{PD}$

but $\frac{RD}{PD} = \frac{RP}{QP}$

$\Rightarrow \frac{RC}{QA} = \frac{RP}{QP}$

$\Rightarrow \frac{RC}{PR} = \frac{QA}{QP} = \frac{1}{2}$ ($A = \text{midpt } PQ$)

$\Rightarrow RC = \frac{1}{2} PR$ as required. ③

Question 9

$$a) \int_{-1}^1 (1-x^4) dx = \left[x - \frac{x^5}{5} \right]_{-1}^1 = 2\left(1 - \frac{1}{5}\right) = \frac{8}{5} \quad \textcircled{3}$$

$$\Rightarrow 1-u^4 = \frac{4}{5} \quad (0 < u < 1)$$

$$\Rightarrow u^4 = \frac{1}{5}$$

$$\Rightarrow u = \sqrt[4]{\frac{1}{5}}$$

$$b) i) \text{Area} = l \times b$$

$$A(u) = 2u \times f(u)$$

$$= 2u \times (1-u^4)$$

$$ii) A(0.5) = 2(0.5) \times (1-0.5^4) = 1 - \frac{1}{16} = \frac{15}{16}$$

$$iii) \frac{dA}{du} = \frac{d}{du} (2u - 2u^5)$$

$$= 2 - 10u^4$$

$$= 0 \text{ for stationary pt}$$

$$\Rightarrow u^4 = \frac{1}{5}$$

$$\Rightarrow u = \sqrt[4]{\frac{1}{5}}$$

$$= 0.66874 \text{ to 5 sig figs}$$

Test nature of SP

$$\frac{d^2A}{du^2} = -40u$$

$$< 0 \text{ for } u = \sqrt[4]{\frac{1}{5}}$$

$$H. \text{ max S.P. at } u_1 = 0.66874 \text{ (sf)}$$

$$iv) A(u_1) = 2\left(\sqrt[4]{\frac{1}{5}}\right)\left(1 - \frac{1}{5}\right)$$

$$= \frac{8}{5} \left(\sqrt[4]{\frac{1}{5}}\right)$$

$$= 1.06998$$

$$= 1.0700 \text{ to 5 sig fig.}$$

$$v) A(u) = uI$$

$$\Rightarrow 2u(1-u^4)^{\frac{1}{4}} = uI$$

$$= u \times \frac{8}{5}$$

$$= u, \text{ (see iv above).} \quad \textcircled{9}$$

Question 10

a) i) After 1 year:

$$A_1 = (P-m)(1+r\%)$$

$$= (P-m)R$$

$$= \text{A (P-m) remaining}$$

ii) After 2 years

$$A_2 = ((P-m)R - m)R$$

$$= PR^2 - mR^2 + mR$$

$$= PR^2 - mR(1+R)$$

iii) A_3 after 3 yrs

$$A_3 = \{[PR^2 - mR(1+R)] - m\}R$$

$$= PR^3 - mR^2(1+R) - mR$$

$$= PR^3 - mR(R^2+1) - mR$$

$$= PR^3 - mR(1+R+R^2) \quad \textcircled{3}$$

$$b) (1+R+R^2)(R-1)$$

$$= R - 1 + R^2 - R + R^3 - R^2$$

$$= R^3 - 1$$

$$\Rightarrow 1+R+R^2 = \frac{R^3-1}{R-1} \quad \textcircled{1}$$

c) Following the pattern in a)

$$A_n = PR^n - mR(1+R+R^2+\dots+R^{n-1})$$

$$= PR^n - mR \frac{(R^n-1)}{R-1} \quad \textcircled{2}$$

d) If $A_n = 0$

$$PR^n = \frac{mR(R^n - 1)}{R - 1}$$

making R^n the subject:

$$\Rightarrow R^n(PR - P) = mR R^n - mR$$

$$\Rightarrow R^n(PR - P - mR) = -mR$$

$$\Rightarrow R^n = \frac{mR}{P - RP + Rm}$$
$$= \frac{mR}{P - R(P - m)} \quad (3)$$

e) $\$P = \$400\,000$ $\$m = \$40\,000$

$$r = 7\%$$

$$\Rightarrow 1.07^n = \frac{40000(1.07)}{400000 - 1.07(360000)}$$

$$\Rightarrow 1.07^n = \frac{42800}{14800}$$
$$= 2.89189...$$

$$\Rightarrow n = \frac{\log(2.89189...)}{\log 1.07}$$
$$= 15.7 \text{ yrs (3 sig fig).}$$

(3)